THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS VII — MATHEMATICS · ANNUAL / FULL-YEAR PRACTICE PAPER

Session 2026-27 | CBSE / RBSE Pattern

Class: 7	Subject: Mathematics	Set 01	Time: 2 Hours 30 Minutes	Max. Marks: 60

General Instructions:

1. This question paper contains 16 questions. All questions are compulsory.
2. Question paper is divided into FIVE sections – Section 'A', 'B', 'C', 'D' and 'E'.
3. In Section 'A' question number 1 has 12 multiple choice sub-parts carrying 1 mark each.
4. In Section 'B' question numbers 2 to 7 are very short answer type questions carrying 2 marks each.
5. In Section 'C' question numbers 8 to 10 are short answer (SA) type questions carrying 3 marks each.
6. In Section 'D' question numbers 11 to 13 are long answer (LA) type questions carrying 5 marks each.
7. In Section 'E' question numbers 14 to 16 are case-based questions whose sub-parts are of 1, 1 and 2 marks respectively.
8. There is no overall choice. However, internal choice has been provided in 1 question of Section B, 1 question of Section C, 2 questions of Section D, and in the 2-mark sub-part of every case-based question in Section E. Attempt only one of the alternatives in all such questions.
9. Draw neat figures wherever required. Use of calculators is NOT allowed. Write the serial number of the question before attempting it.

SECTION A — Multiple Choice Questions (12 × 1 = 12) — 1 questions · 12 marks

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) One billion is equal to how many crore? **1**

(a) 100 crore (b) 10 crore (c) 1000 crore (d) 1 crore

(ii) The number of seconds in one day is: **1**

(a) 14,400 (b) 3,600 (c) 8,640 (d) 86,400

(iii) The value of 36 ÷ 6 × 3 + 2 is: **1**

(a) 4 (b) 30 (c) 20 (d) 8

(iv) In the decimal number 4.058, the digit 5 is in the: **1**

(a) hundredths place (b) tenths place
(c) thousandths place (d) tens place

(v) Which of these decimals is the greatest? **1**

(a) 0.49 (b) 0.499 (c) 0.05 (d) 0.5

(vi) 'Five more than three times a number n' is written as: **1**

(a) 5n + 3 (b) 3(n + 5) (c) 3n + 5 (d) 3n − 5

(vii) The value of $2a^2 - 3b$ when a = 3 and b = 2 is: **1**

(a) 6 (b) 0 (c) 12 (d) 24

(viii) One angle of a linear pair is 65°. The other angle is: **1**

(a) 25° (b) 125° (c) 295° (d) 115°

(ix) A transversal cuts two parallel lines. Each pair of co-interior angles (interior angles on the same side) adds up to: **1**

(a) 180° (b) 90°
(c) 360° (d) a different value each time

(x) The sum of the first 10 odd numbers is: **1**

(a) 55 (b) 110 (c) 90 (d) 100

(xi) Which three lengths can be the sides of a triangle? **1**

(a) 4 cm, 5 cm, 9 cm (b) 7 cm, 8 cm, 10 cm

(c) 3 cm, 3 cm, 7 cm (d) 6 cm, 6 cm, 13 cm

(xii) The value of 2/3 ÷ 4/9 is: **1**

(a) 3/2 (b) 8/27 (c) 4/3 (d) 1/6

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Write 5 crore 8 lakh 40 thousand in figures using commas (i) in the Indian system and (ii) in the international system. **2**

3. Evaluate: 48 ÷ (4 + 2 × 2) + 3 × 2. Show each step. **2**

4. A ribbon 3.5 m long is cut into pieces of 0.25 m each. How many pieces are obtained? **2**

OR

Subtract 4.875 from 10.2.

5. Simplify: 5x + 3y − 2x + 4y. **2**

6. Two straight lines intersect and one of the four angles formed is 40°. Find the other three angles, giving reasons. **2**

7. Start with 3524 and carry out the Kaprekar routine (arrange digits in descending and ascending order and subtract) until you reach 6174. **2**

SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. In triangle ABC the angles are in the ratio ∠A : ∠B : ∠C = 2 : 3 : 4. Find the three angles and state whether the triangle is acute-angled, right-angled or obtuse-angled. **3**

OR

The exterior angle of a triangle is 110° and one of its interior opposite angles is 45°. Find the other interior opposite angle and the adjacent interior angle.

9. Meena's mother needs 3/4 cup of flour for one batch of rotis. She has 4 1/2 cups of flour. (i) How many batches can she make? (ii) She also uses 2/3 of a 9/10 kg pack of ghee; how many kg is that? **3**

10. A city has 12,50,000 houses and every house uses about 480 litres of water a day. Find the total water used in a day (i) in litres, written in the Indian system and in crore, and (ii) in the international system (in millions). **3**

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

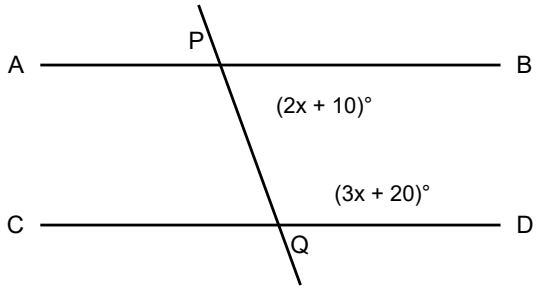
Questions 11 to 13 carry 5 marks each.

11. Sunita buys 2.5 kg rice at ₹ 48.60 per kg and 1.25 kg sugar at ₹ 44.80 per kg. (i) Find the cost of the rice. (ii) Find the cost of the sugar. (iii) Find the total bill. (iv) She pays with a ₹ 200 note; how much change does she get? (v) How many 50-paise coins make up the cost of the rice? **5**

OR

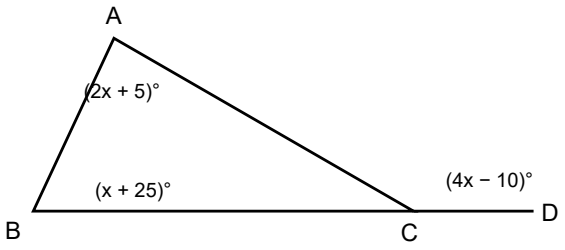
(i) Write 7/8 as a decimal. (ii) Write 0.375 as a fraction in simplest form. (iii) Arrange 0.6, 0.59, 0.605 and 0.06 in ascending order. (iv) Find 4.56 × 100. (v) Find 7.5 ÷ 10.

12. In the figure, AB ∥ CD and PQ is a transversal. ∠BPQ = (2x + 10)° and ∠DQP = (3x + 20)°. (i) Name the relation between these two angles. (ii) Find x. (iii) Find ∠BPQ and ∠DQP. (iv) Find ∠APQ. **5**



OR

In the figure, side BC of triangle ABC is extended to D. ∠A = (2x + 5)°, ∠B = (x + 25)° and the exterior angle ∠ACD = (4x − 10)°. (i) State the exterior angle property. (ii) Find x. (iii) Find ∠A and ∠B. (iv) Find ∠ACB.



13. Squares are made in a row using matchsticks: 1 square needs 4 sticks, 2 squares need 7 sticks, 3 squares need 10 sticks. (i) How many sticks are needed for 5 squares? (ii) Write a rule (expression) for n squares. (iii) How many squares can be made with exactly 100 sticks? (iv) Can exactly 50 sticks make a row of squares with none left over? Why? (v) Is the number of sticks for n squares always odd? Check for n = 1, 2, 3, 4. **5**

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. Rajasthan is the largest state of India by area. According to the 2011 Census, its population was 6,85,48,437 and its area is 3,42,239 square kilometres.

(i) Write the population correct to the nearest lakh. 1

(ii) How many crore people is this, approximately (to two decimal places)? 1

(iii) Find the approximate population per square kilometre (nearest whole number). 2

OR

(iii) If the population increases by 25,00,000, write the new population in the Indian system.

15. A triangular flower bed ABC in a school garden has AB = 9 m and BC = 7 m. The third side CA = x metres is a whole number. The angles at A and B are 50° and 70°. The gardener extends side BC beyond C to lay a path.

(i) Can CA be 17 m? Give a reason. 1

(ii) Write the greatest and the smallest whole number value of x. 1

(iii) Find ∠C and the exterior angle at C. 2

OR

(iii) If a similar flower bed is isosceles with a vertex angle of 40°, find its base angles.

16. Kavya's mother has 12 1/2 litres of milk in a can. She pours it into glasses, each holding 1/4 litre.

(i) How many glasses can she fill? 1

(ii) Write the reciprocal of 5/8. 1

(iii) She sells 3/5 of the milk. How many litres are left? 2

OR

(iii) Each glass is sold at ₹ 15. Find the money received if 3/5 of the milk is sold.